

1. BACKGROUND

The lack of continuous, molecular level insight into the body, for example real-time tracking of hormones, cytokines, and drug levels, leads to preventable disease progression and missed chances for early intervention. Wearable technologies could change this by providing timely data to inform a specific course of action or modify behavior in real time. Unfortunately, current devices are often cumbersome, inaccurate, or too costly because each biosensor is engineered as a bespoke system with expensive, non-reusable components.

Delphi will improve the biosensor development ecosystem, enabling low-cost but high performance sensing by shifting to a chiplet-based architecture. With chiplets, components of what would typically be an Application Specific Integrated Circuit (ASIC) can be reused and remixed into new devices that share the same high performance. Yields and scaling tend to be easier, and the approach inherently enables heterogeneous integration, which is ideal due to the diverse material set often needed for biosensing. Delphi aims to leverage existing advances in chiplets to focus on the components specific to biosensing. First, dry, hermetically sealed chiplets must provide ultra-low power management, overcoming the limitations of current power-hungry standards and the constraints of low duty cycle operation and RF or internal power harvesting. Additionally, chiplets that ensure secure communication of medical data from these body-worn devices must be developed. Second, “wet” biosensor chiplets that directly interface with biology are needed. Delphi aims to extend the state of art from biophysical measurements or high-concentration metabolites to the continuous measurement of low concentration biochemical markers such as hormones and cytokines over extended periods. Third, these chiplets must be integrated into biocompatible packages that are fully encapsulated except at the sensing interface. Existing encapsulation methods either lack the selectivity and reliability required at the chiplet scale or depend on rigid, complex, non-modular solutions. New packaging strategies must protect delicate electronics, preserve sensor surface chemistry, and support durable, long-term operation in biological environments.

By the end, Delphi will quickly and cheaply develop modular wearable or ingestible sensors for monitoring a broad range of biological markers.

2. PROGRAM OVERVIEW

Today, the lack of deep biological insight — such as continuous monitoring of hormones, cytokines, and drug levels — contributes to preventable disease progression, missed opportunities for early intervention, and limited personal health agency. Emerging wearable technologies could close this gap by providing real-time physiological data, enabling individuals and clinicians to recognize early warning signs, tailor treatments, and adjust behaviors in the moment rather than after damage has occurred. Consequently, Delphi seeks to empower all Americans to fully understand their body by providing this information. This will require a revolution in the cost and performance of biosensors. The program will leverage chiplets, a new system architecture in microelectronics, to greatly decrease cost, reduce size,

weight, and power (SWaP), and enable all developers to access cutting edge capabilities. By the end, Delphi will quickly and cheaply develop modular wearable or ingestible sensors for monitoring of a broad range of biological markers. The benefits include early detection to prevent disease progression, continuous monitoring for safe home recovery, and precise, closed-loop therapy for superior health outcomes.

PROGRAM DESCRIPTION

The Delphi program is soliciting proposals to create an open, extensible platform to support wearable and ingestible biosensors. Current bioelectronic devices are largely built from scratch for each application, using discrete components on printed circuit boards and, only later, costly custom ASICs. Because each ASIC must integrate multiple functions onto a single die, every design becomes its own expensive, time-consuming project, slowing progress and limiting performance. Delphi performers will create a standard format of packaging and interconnections that will enable integration of chipelets from multiple manufacturers. **Proposers should identify preferred packaging and interconnect solutions; however, a series of meetings in the first 2 months of the program will establish a consensus packaging approach, thus ensuring interoperability.** Given the needs of biosensing, ultra-high density interconnects are unlikely to be needed. Approaches that integrate power, such as a battery or an RF antenna, into the substrate are encouraged.

Teams will likely need to develop Delphi specific “dry” chipelets for capabilities unique to biosensing (e.g. power management and highly secure communications) but are expected to leverage commercially available chipelets or IP for more generic device functions, thereby leveraging the cost savings due to production scales. While development and testing of implantable devices is out of scope for the Delphi program, platforms that could fit within the size, weight, and power limitations of an implantable device or serve as wearable or ingestible would be viewed favorably.

Delphi teams will develop “wet” biosensing chipelets capable of real-time measurement of deep biological information. Except for continuous glucose monitors, most commercial sensors measure only biophysical signals. Routine, continuous detection of low concentration biochemical analytes, such as hormones (0.1 – 100 µg/L) and cytokines (~10 ng/L), has been demonstrated by only a few groups, and typically for just a few days in wearable formats. New sensing modalities demonstrate the potential for longer term, regenerative, measurements that could be adapted to hormones, cytokines, or other markers. However, each new biomarker currently demands a dedicated discovery and optimization process along with device re-engineering, limiting scalability and slowing the development of the multiplexed sensor panels needed for comprehensive health profiling. Delphi is seeking approaches that are generalizable to multiple analytes to increase the extensibility of sensor systems and platforms. Additionally, the development of surface chemistries that prevent sensor fouling is typically arduous. Delphi is open to new concepts that can rapidly improve the passivation of sensor surfaces to deliver methods that are compatible with clean rooms. Performing sensor functionalization within the fab, likely in a separate post-processing hood, would minimize the surface contamination that hinders the formation of high-quality passivation. To meet these challenges, teams must develop design, optimization, and manufacturing processes that support rapid retargeting of new

analytes and scalable production, including chiplet printing and processing, device packaging, and sterilization.

Delphi teams will need to develop robust biocompatible materials and encapsulation processes for protecting electronics, while providing biosensors access to their target environment. Conventional materials and fabrication methods often fall short when applied to small electronic components. These conventional approaches frequently result in leaks or failures, as they are not designed to accommodate the unique challenges posed by miniaturized devices. Moreover, there is currently no solution that can simultaneously protect delicate electronics, permit selective exposure where needed, and preserve the surface chemistry essential for sensor function. Approaches must be made compatible with the underlying biochemistry of the biosensor chips while meeting the extremely tight tolerances and critical materials properties associated with chiplet fabrication to prevent, for instance, warping of the small system.

NATIONAL HEALTH IMPACT

A national ARPA-H program to enable small, low-cost, high-performance biosensors would address key gaps: the lack of scalable platforms for deep molecular monitoring outside specialty centers; the need for minimally invasive, high-frequency data on inflammation and metabolic status; and technical barriers in miniaturization, power, and multimodal sensing. Emerging technologies — such as self-powered ingestible metabolite sensors, integrated GI smart capsules, interstitial fluid-sampling microneedles, and sweat-based biosensors — demonstrate that battery-free or low-power operation, multi-marker detection, and wireless telemetry are feasible in preclinical and early clinical settings. If these platforms can be engineered for low-cost, robust manufacturing and designed for usability and connectivity in resource-limited environments, they could expand access to high-frequency molecular data beyond infrequent lab testing, enabling earlier detection of disease activity and more adaptive management.

Delphi seeks to lower the cost of all biosensors and is agnostic to the targeted disease.

However, proposers must identify their targeted disease or health or wellness indications and justify their choice, as described below. Examples of benefits to patients could include earlier detection of flares in inflammatory bowel disease and rheumatoid arthritis, pre-symptomatic recognition of sepsis or pulmonary hypertension progression, and dynamic monitoring of treatment response, particularly in rural and low-resource settings. By bending the cost curve, Delphi hopes to vastly increase the market for biosensors, enabling individuals to better understand and improve their own health thereby reducing expensive and avoidable hospitalizations and emergency visits.

PROGRAM SCOPE

Delphi will unite the scalable production of commercial microelectronics with advances in biosensor detection and biocompatibility coming out of academia and other research institutions. Multidisciplinary teams must include at least three key roles: (1) a chiplet group (interposer/substrate, interconnects, power management, secure communication), (2) a “wet” biosensor group for detecting diverse humoral biomarkers, and (3) an encapsulation group ensuring hermetic sealing while exposing the biosensing interface to biological

environments. **Teams, not ARPA-H, are responsible for full integration. Proposals that do not address all TAs and their integration into a functional device will not be considered.**

Proposers should choose one of two tracks: (1) Minimally invasive devices that either penetrate the stratum corneum or are ingested. These devices require FDA oversight and will culminate in formal human clinical trials to validate safety and efficacy. (2) Non-invasive wearables that do not break the skin's surface which aim to promote healthy lifestyle choices and reduce disease risk. These devices have a lower regulatory barrier and will culminate in human factor testing to assess usability and impact. This dual-track approach supports both advanced medical devices and wellness-focused wearables.

3. DELPHI TECHNICAL AREAS

Delphi will quickly advance the development of modular, chiplet-based wearables and ingestibles to monitor a wide range of biological markers. Current bioelectronic devices are custom-built for each application, relying on discrete components in relatively large PCB boards or smaller ASICs that integrate multiple functions onto a single die, but that are costly to develop and to further refine. A chiplets architecture enables both smaller and high-performance systems while lowering the cost of development and ongoing edits. Delphi aims to leverage existing capabilities in the chiplet space while using the Technical Areas (TAs) defined below to focus on the capabilities unique to biosensing. To this end, Delphi comprises four interconnected TAs, all of which must be addressed:

TA1 $\equiv$ Dry Chiplets Unique to Biosensing: Determine a common packaging and interconnect strategy. Develop chiplets for ultra-low power management and for secure medical data transmission.

TA2 $\equiv$ Wet Biosensing Chiplet: Develop biology-facing sensors for detection of deep biological data that interfaces into the chiplet architecture. This will include not only improvements in biological recognition elements but new approaches for surface passivation that are compatible with clean room processing.

TA3 $\equiv$ Biocompatible Encapsulation: Develop hermetic encapsulation to seal integrated chiplet packages, exposing the “wet” sensing interface to the biological environment while preserving the “dry” chiplets from moisture ingress.

TA4 $\equiv$ Human Factors Testing or Clinical Trial: Based on the track chosen, assess whether the device can be used safely, effectively, and intuitively by intended users under realistic conditions, identifying and mitigating any use-related risks.

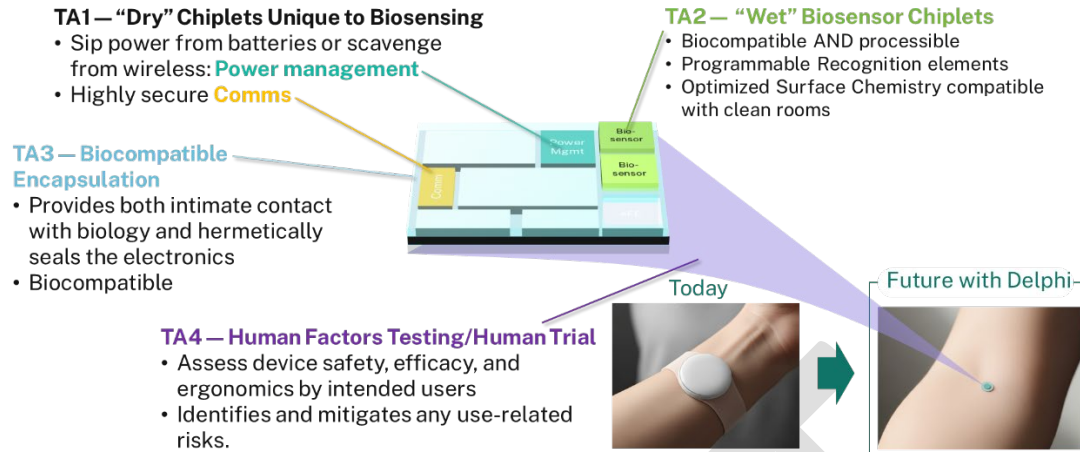


Figure 1. Summary of Program Technical Areas

TA 1: “DRY” CHIPLETS UNIQUE TO BIOSENSING THAT ARE HERMETIC AND WILL SERVE AS THE FOUNDATION FOR THE BIOSENSING CHIPLET COMPONENTS INTEGRATION

Under TA1, proposals should present a scalable and cost-effective strategy for packaging chiplets for biosensing. To ensure interoperability, a consensus approach will be agreed upon in the first two months of the program, beginning with initial discussion at the program kickoff. This foundational package includes the chosen substrates, interconnects, and die-to-die interface. Moreover, biosensing presents different requirements than much of the current work in chiplet design, so the expectation is that at least two new chiplets must be designed including one to optimize power management via battery or wireless power harvesting and one to securely, ideally post-quantum secure, transmit medical information to a handheld device. TA1 aims to create a **standard** chiplet foundation that is modular and interoperable across the different teams. Indeed, a goal of Delphi is to establish a flexible ecosystem for developing advanced biosensors. To this end, interoperability will be tested in all phases of the effort.

Multiple concurrent objectives will be pursued.

Objective 1: Develop Optimal Packaging and Interconnect Solutions Using Industry Standards

For packaging, proposals should present a scalable and cost-effective strategy for packaging chiplets for biosensing. The expectation is that companies with established expertise in electronics design, manufacturing, and packaging, or academic institutions closely partnered with such industry leaders will develop optimal packaging and interconnect solutions. A strategy for die-to-die connectivity, such as UCIe, should be proposed. Any necessary deviations from an existing standard should be clearly identified. Additionally, any existing chiplets that can be reused or effectively repurposed for biosensing and with the proposed packaging should be identified. The goal is to enable

seamless interoperability of off-the-shelf chiplets from diverse vendors, ensuring absolute compatibility across components and importantly, *across the different teams*.

Once the initial proof-of-concept integrated dry chiplet package is established, teams will demonstrate that their solutions are interoperable with the other the TA1 from other teams. Each team should carefully consider the co-engineering constraints. For instance, choice of packaging could impact the ability to functionalize sensors in TA2 and the ability to encapsulate the device in TA3. Teams should present a coherent strategy that ensures robust electronic performance, high-sensitivity detection, and long-term biocompatibility.

Objective 2: Develop Chiplets for Secure Communication Chiplets

A successful program will lead to the widespread monitoring and broadcasting of sensitive medical information. Performers from each team will either design or select commercially available chiplets that can communicate securely, ideally with post-quantum communication. These chiplets will be integrated into the chosen architecture to ensure robust protection of sensitive medical and biosensing data exchanged within the chiplet ecosystem. Chiplet-based communications must maintain the confidentiality, integrity, and authenticity of sensitive medical data even against quantum-capable attacks by employing energy-efficient post-quantum cryptographic mechanisms for the devices. The objective is to future-proof the architecture against emerging security threats and guarantee compliance with advanced cryptographic standards. Notably, security must also account for the low amount of power that will be available.

Objective 3: Advance Power Management Solutions for Biosensing Applications

Compared to other electronic systems, biosensors tend to remain in a dormant state for extended periods, waking to make a measurement and to broadcast the results. This modality brings greater importance to energy consumption during the sleeping state in which the system typically finds itself. Additionally, miniaturization increases the importance of leveraging either the power available from a minute battery or from scavenging energy. For this objective, teams should develop a chiplet that can resolve these demanding constraints. Repurposing existing chiplets that meet these stringent requirements is also allowed. Teams should describe how changes to the frequency and modality of sampling could influence power management strategy. **Proposals must include an estimated power budget for their system and clearly define how it would be managed by the proposed chiplet.** The goal is to maximize energy efficiency, enable ultra-low power operation, and/or ensure robust power harvesting to support reliable device performance for one week.

Objective 4: Establish Integrated Dry Chiplet Packages and Interoperability

In Phase I, teams will assemble and validate the first integrated dry chiplet package, demonstrating the successful combination of substrates, interconnects, communication, and power management components. Using the agreed common framework, each team will demonstrate interoperability and data exchange with chiplets developed by other teams. This sequence of development described in Table 1. *Proprietary protocols are highly discouraged*. To incentivize interoperability, during Phases II and III, there will be “remixes” of components to generate new sensors as defined by the ARPA-H program manager. In this scenario, use of other team’s proven solutions, rather than redevelopment, would be highly efficient. This proof-of-concept will serve as a benchmark for modularity, reusability, and

performance, setting the stage for subsequent integration with biosensing and encapsulation technologies.

TA 2: “WET” BIOSENSING CHIPLETS THAT ARE MODULAR, INTERCHANGEABLE ON THE MAIN CHIPLET SUBSTRATE AND SPECIFIC FOR THE SELECTED USE CASE.

Beyond leveraging chiplets to enhance bioelectronics, a key goal is to integrate and expand recent advances in the detection of “deep” biological information, lower concentration analytes that provide key information about the status of the body. To this end, under TA2, proposals should detail the creation of biosensing chiplets that can continuously track hormones, cytokines, or drug levels. “Continuously” is defined by default as the Nyquist frequency for the targeted soluble factor. Teams may propose other sampling frequencies with justification of how they will meet the goal of perceived real-time monitoring. To meet these ends, the wet biosensing chiplet should be engineered with durable surface chemistries capable of long-term operation in biological environments and optimized to detect analytes at appropriate duty cycles.

Proposers must identify the primary use case of interest, including the disease and the biomarker(s) they intend to detect, the sensing modality, and the frequency of sensing for development and integration with the TA1 chiplet. For example, teams wanting to measure a cytokine must determine if a wearable or an ingestible sensor is appropriate, the sample medium, as well as the sampling frequency that provides the optimal status of the physiological state. These critical design choices will inform SWaP determinations for TA 1 and should be determined early in the process.

Reproducible surface chemistry is at the heart of robust sensing. Proposed work should account for 1) the chiplet must be sensitive and selective while avoiding degradation and fouling, 2) regeneration of the molecular probes to avoid saturation or some other robust technique to enable continuous sensing, and 3) surface functionalization compatible with clean room processing. Depending on the sensor transduction strategy, the biosensor chemistry must also support selective encapsulation, ensuring the cleanliness of the area to be functionalized hermetically sealing the electronics. In the remixes, teams will be required to develop three different sensors targeting different analytes, one in each of the program phases. Consequently, the strategy for molecular recognition should be flexible.

Objective 1: Achieve high sensitivity and selectivity while avoiding biosensor degradation and fouling.

Teams must design sensors with limits of detection appropriate to the specific sensing modality (e.g., ingestible, wearable, ISF), ensuring they can reliably detect physiologically meaningful analyte concentrations in different states (healthy vs. disease). Teams will demonstrate the sensor is highly specific for the intended analyte to avoid cross-reactivity and interference from other biomolecules present. During the program, performers will be required to demonstrate sensitivity at various levels of detection, culminating in sensors that can reproducibly detect the physiologically relevant levels of the target analyte *in vivo*.

Surface passivation of sensors to prevent fouling tends to be time consuming. Proposers should clearly state their chosen chemistry, its appropriateness to the task, and how well developed it has been in their hands (e.g., samples in buffer, spiked clinical samples, use *in*

vivo, etc.). Approaches that enable more rapid development, such as AI-driven experimentation, are also of interest.

Objective 2: Demonstrate probe regeneration to avoid saturation.

Sensor specificity often comes at the cost of tight binding of the recognition element which then degrades in time. Since Delphi biosensors will be used across multiple sampling events, surface chemistries that regenerate probes to avoid saturation are that otherwise maintain accurate continuous sensing should be proposed. Throughout the program, teams will demonstrate functionality and increasing levels of sensitivity across *in vitro* and *in vivo* settings.

Objective 3: Fabrication with materials compatible with both biology and microelectronics and under clean room conditions.

Successful integration with TA1 and TA3 efforts requires biosensors constructed with materials compatible with biology, containing the recognition elements, and with microelectronics for integration with the TA1 architecture. Final production in clean rooms is required to ensure integration with TA1 chiplets and proper encapsulation under TA3.

Objective 4: Demonstrate extensibility of the sensing technology

As the program progresses, teams will be required to adapt their sensor technology for detection of additional target molecules of interest. Proposals should describe how their sensor technology can be adapted for sensitive and specific detection of new targets. As described in the program structure below, in Phase II teams will be required to demonstrate detection of a second biomolecule of interest and in Phase III a third. In the proposal, teams must propose at least 5 options from which the Government will select the second target at the beginning of Phase II. The Government will share the third target at the beginning of Phase III from options that are outside the original list.

TA 3: BIOCOMPATIBLE ENCAPSULATION TO SEAL INTEGRATED CHIPLETS AND ALLOW THE SENSING PORTION TO REMAIN EXPOSED TO THE BIOLOGICAL ENVIRONMENT

Proposals should describe the proposed solutions for encapsulation that protects sensitive electronics from moisture ingress, selectively exposes the sensor as needed, and does not elicit an immune response. The vision is of a complete device assembled from the dry and wet chiplets developed in other TAs and then hermetically encapsulated except at the “wet” interface to the biological environment, ensuring biocompatibility and long-term reliability in their intended application. Consequently, team members will collaborate closely to ensure seamless integration. We note the challenge of encapsulating the system without contaminating the surfaces whose cleanliness are critical for success in TA2. Consequently, proposers should clearly explain their strategy for selective sealing of biosensing chiplets while guaranteeing full protection from biological fluids. Moreover, TA3 must demonstrate that the devised coating strategies are completely biocompatible, chemically inert, resistant to accelerated degradation, and unaffected by common sterilization processes such as ethylene oxide or gamma irradiation.

Objective 1: Achieve Advanced Hermetic Sealing for Component-Level Chiplets

Collaborate with TA1 to develop and implement encapsulation strategies that meet the highest standards of hermetic sealing (MIL-STD-750 (1071) for discrete component

hermeticity and MIL-STD-883 (1014) for the complete electronic packages, or comparables), ensuring robust protection of chiplets from environmental and biological exposure.

Objective 2: Enable Selective Sealing for Biosensing Chiplet

Work closely with other team members working on TA2 to devise selective sealing methods that allow biosensing chiplets to interface directly with biological environments, while safeguarding microelectronics from any biological fluids. This ensures absolute biocompatibility and supports accurate, long-term sensor readings.

Objective 3: Demonstrate Superior Coating Performance

Validate that all coatings used are fully biocompatible, chemically inert, resistant to accelerated degradation, and impervious to common sterilization methods such as ethylene oxide and gamma irradiation, guaranteeing durability and safety for medical applications. Proposers should clearly state the intended assembly, sterilization, and application process.

TA 4: HUMAN FACTORS TESTING OR CLINICAL TRIAL

TA4 proposals should articulate a comprehensive plan for human factors testing to be conducted once advanced, functional prototypes are developed. Testing will follow two distinct tracks: non-invasive and minimally invasive. The non-invasive track centers on wearables that do not breach the skin's surface, targeting the promotion of healthy lifestyle behaviors and the reduction of disease risk. Devices in this category are not regulated by the FDA, and human factors testing at the conclusion of development will evaluate usability, user experience, and behavioral impact.

The minimally invasive track encompasses devices designed to penetrate the skin or be ingested, aligning with traditional medical device pathways. These devices are subject to FDA oversight and will undergo rigorous human clinical trials to confirm safety and efficacy. By structuring the program around these two tracks, TA4 ensures that both wellness-oriented wearables and advanced medical devices are addressed, broadening the scope and impact of the initiative.

Objective 1: Non-invasive human factors evaluation

For the non-invasive track, the primary objective is to thoroughly evaluate the usability and user experience of wearable prototypes through comprehensive human factors testing. This process will focus on their ergonomics and how effectively these devices encourage healthy behaviors in everyday settings. By collecting detailed user feedback, the development team will refine device design, comfort, and engagement, ensuring that the wearables are widely adopted and used consistently. The goal is to demonstrate that these devices can successfully promote lifestyle changes without requiring FDA regulation. Proposers are strongly encouraged to read the most recent FDA guidance to ensure that their product falls within this category.

Objective 2: Minimally-invasive human clinical trials

In the minimally invasive track, the objective shifts to validating the safety and efficacy of prototypes that penetrate the skin or are ingested. These devices will undergo formal review to ensure compliance with FDA requirements and regulatory standards for medical devices. The human clinical trials will assess safety, clinical performance, reliability, and biocompatibility within target populations, generating robust evidence to support

regulatory approval and integration into medical practice. This approach ensures that minimally invasive devices are not only safe and effective but also ready for widespread adoption in healthcare settings. ARPA-H will only support efforts through Phase I clinical trials.

PROJECT MANAGEMENT AND INTEGRATION

Proposals should clearly describe plans for overall project management and for integration across technical areas. This should include not only plans for integration and manufacture of prototype and final devices, but strategies for ongoing integration to ensure lab scale and manufacturing scale processes are aligned towards optimal compatibility and rapid translation. The person or people directly leading integration should be identified as well as the hours they will dedicate to integration each year.

PROGRAM STRUCTURE

Delphi is a 54-month, 3-phase program that consists of four TAs. During Phase I (24 months), performers will conduct component and process development including wet and dry chiplet development. During Phase II (18 months), performers will integrate chiplets and validate functionality *in vivo*, while also developing any additional chiplets needed for the first remix. During Phase III (12 months), performers will scale for manufacturing, develop any chiplets needed for second remix (ideally none), and conduct human factors testing or a human clinical trial. Performers will produce multiple biosensor chiplets across the three phases through remixes to ensure modularity and reusability of the ecosystem.

Phase I (24 months): Performers will conduct component and process development. Teams will demonstrate *in vitro* and *in vivo* testing of their first device. Under TA1, performers will identify common standards and will develop the underlying platform, dry chiplet, and the first device. For TA2, teams will develop the first wet chiplet for the first device. For TA3, teams will develop new materials and demonstrate compatibility with manufacturing and sterilization. For TA4, teams will conduct risk analysis and prepare for regulatory engagement.

Phase II (18 months). Integration and Preclinical. Teams will demonstrate functional capability of the fully assembled first device *in vivo*. **Appropriateness of chosen animal model should be clearly stated in the proposal.** For TA1, performers will assemble the first device. For TA2, teams will develop a second biosensor chiplet for a different molecule of interest. For TA3, teams will demonstrate integration and adherence to standards. For TA4, teams will conduct regulatory engagement to prep for Phase III trials.

Phase III (12 months). Clinical. For TA1, teams will perform a second device remix and scale manufacturing for testing. For TA2, teams will develop a third sensor for a different biomolecule of interest. As needed, team members working on TA3 will continue demonstrating integration and adherence to standards; however, the expectation is that processing should be well-developed by this point. TA4 teams will conduct human factors testing or a human trial depending on the track chosen.

PROPOSAL SCOPE

To be responsive to this Solicitation, proposals must address all four TAs (TA1, TA2, TA3, & TA4). Proposals not addressing all four will not be reviewed. An organization may serve as a sub-performer under multiple proposals. If ARPA-H selects two or more proposals in which the same sub-performer is proposed to perform work under the same Technical Area, ARPA-H reserves the sole right to determine whether the proposed efforts are substantially distinct or whether a material overlap exists. In the event ARPA-H determines that such an overlap exists, ARPA-H may, at its discretion, (i) fund the overlapping work only once, or (ii) require the sub-performer to elect the project in which it will participate. Proposers must elect TA4 track 1 (Non-invasive) or TA4 track 2 (minimally invasive). Developing either of these integrated devices will require performers to address all applicable TAs.

Proposals are expected to include all the required demonstrable expertise to achieve the goals of all four TAs. Specific content, communications, networking, and team formation are the sole responsibility of the proposing teams. **Proposers must submit a single proposal led by one PI under a single prime performer that addresses program Phase I, Phase II, and Phase III.**

Tracks

Non-invasive: Proposals with methodologies considered non-invasive by FDA standards will undergo Human Factors testing for Phase III.

- Examples include a sweat sensor or photoacoustic interrogation of blood.
- No FDA interaction is necessary as these sensors are expected to “promote, track, or encourage choices that, as part of a healthy lifestyle may reduce the risk of certain chronic diseases or help living well within certain chronic disease conditions.”

Minimally Invasive: Proposals with methodologies deemed minimally invasive/invasive by FDA standard, yielding a device that penetrates the skin or is ingested, will undergo a clinical trial for Phase III. Proposals under this track must:

- Clearly define the target biomarker and justify choice of indication.
- Engage with the FDA.
- Obtain all necessary regulatory approvals required to execute a human clinical trial.

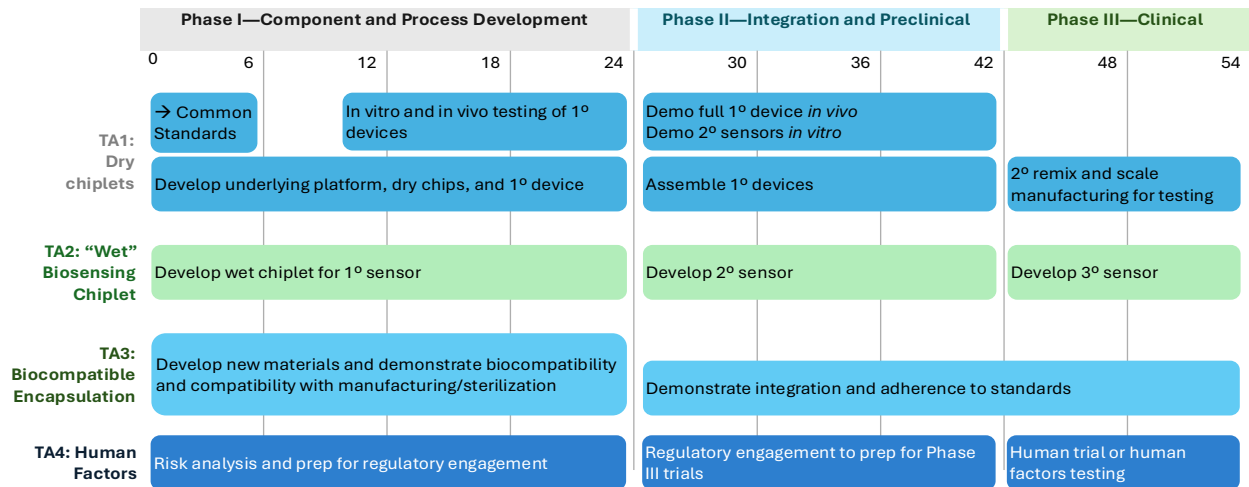
Program Milestones and Metrics

Successful proposals will describe a clear plan to meet these milestones in the program. Phase milestones should be met in all TAs. Proposals should include milestones for their biosensing approach. Note: ARPA-H will meet with Delphi performers at least monthly to review progress towards the milestones and metrics defined below.

An expected schedule for key engineering and regulatory events is below. Performers may propose an alternate timeline with justification.

Program Key Milestones

To ensure modularity and reusability of the ecosystem, two remix challenges will be run



The program metrics below are *minimum* requirements appropriate for all proposers and serve to bound scope of the effort while still affording maximal flexibility, creativity, and innovation of the proposed solutions. **Proposers should propose additional milestones and quantitative metrics appropriate to their specific approach for each Phase of the program.** Achievement of all metrics, as agreed to by ARPA-H, is the basis for initiation of the optional Phases. *Please note that performers are expected to demonstrate efforts towards meeting all relevant metrics. Unsuccessfully addressing a metric may result in the termination of the award.*

APPENDIX A: PROGRAM AND TECHNICAL DESCRIPTION

Table 1. Delphi Program Metrics

Month	TA1—Dry Chiplets	TA2—Wet Biosensing Chiplets	TA3—Biocompatible Encapsulation	TA4—Human Factors
1	Virtual meeting; I&I counterproposals due; Identify any necessary deviations from Open Domain-Specific Architecture			
2	Common I&I chosen			
6			Initiate <i>in vitro</i> biocompatibility studies, if needed	
12	Integration substrate swap to ensure interoperability	Biosensor chiplet LoD <10x physiologically relevant level (PRL)	Initiate accelerated degradation studies of the encapsulation in simulated physiological conditions per ISO 10993-1	
18	Demonstrate adherence to NIST secure communication standards FIPS 140-3 and SP 800-121	Biosensor LoD < PRL in spiked clinical samples. Functional for 7 days in simulated media, sensitivity and specificity drifts over time ≤10%.	• Demonstrate that chosen sterilization approach does not degrade chiplet or interconnect performance • Complete <i>in vivo</i> biocompatibility studies, if applicable, with histological analysis for fibrosis and inflammation • Submit to ARPA-H the intended Biological Evaluation Plan	Complete risk analysis, identifying challenges in sensor placement, operation, and data interpretation
24	• Power management meets specs for one week of continuous operation • Necessary new chiplets for the first remix (2^o) identified	Prototype device demonstration <i>in vivo</i>	Submit to ARPA-H the intended hermeticity evaluation plan per MIL-STD-883 or comparable	Submit to ARPA-H the intended architecture and specs evaluation plan per IEEE 360-2022 and other ANSI/CTA relevant standards
30	• Demonstrate interoperability and data exchange protocols with chiplet from a different team • Demonstrate adherence to MIL-STD-750(1071) or comparable discrete component hermeticity	Assembled primary (1 ^o) device meets FDA guidance for biosensors <i>in vivo</i> : • Accuracy: within ±15% of nominal conc. • Precision: ≤15% coefficient of variation		
36		2 ^o sensor chiplet LoD < 10x PRL		For min. invasive, GMP manufacture and regulatory clearance for human use
42	• Power for 7 days through battery or other source • 2^o sensor fabricated reusing at least 40% of Phase I chiplets • Identify necessary new chiplets for the second remix (3^o)	• Assembled 1^o device has ≤10% drift @ 7 days <i>in vivo</i> • 2^o sensor chiplet LoD <PRL	• Demonstrate adherence to MIL-STD-883 (1014) (or comparable) to hermeticity of electronic packages • Demonstrate adherence to ISO 10993-1 and relevant parts biocompatibility standard	Demonstrate adherence to IEEE 360-2022, architecture and specs of consumer electronic wearables and any additional ANSI/CTA standards
54	3 ^o device fabricated reusing >80% of Phase I and Phase II chiplets	3 ^o sensor chiplet LoD < PRL		Completed Trial or Human Factors Testing

Collaboration and Data Sharing

All components developed under the TA's must be integrated into a single chiplet system that delivers secure and accurate biomarker tracking. As noted above, proposals must identify a team member as the primary integrator of the different TA components; this individual does not need to be the Principal Investigator (PI). Delphi's focus is to create a common chiplet ecosystem, therefore, it is expected that all performers will interact and work collaboratively with other performers in developing the methods, technologies, and tools using open, timely, and effective communication, information exchange, and

reporting. Performers across all partner organizations will attend common meetings and technical exchanges to advance relevant technologies, bridge across data siloes, and move toward a common care delivery platform across numerous clinical use cases.

To facilitate the open exchange of information described above, **performers will have Associate Performer Agreement (APA) language included in their award.** Each performer will work with other Delphi performers to develop an APA that specifies the types of information that will be freely shared across performer teams. The open exchange of scientific information will be critical in advancing the TA1 chiplet hardware ecosystem required to achieve the Delphi objectives. The APA will establish a common understanding of expectations to guide the open exchange of ideas and establish a collaborative foundation for the Delphi program. Each performer will work with other performers as described in the Program Collaboration Requirements (Figure 5).

TA1

With TA 1: TA1 sub-performers will collaborate with TA1 sub-performers from other teams to determine common packaging and interconnect strategies, jointly selecting the relevant approaches and standards. They will exchange their proposed dry chiplet solutions to verify full interoperability and demonstrate compliance with the agreed-upon specifications. The primary objective is to validate interoperability and data exchange protocols between chiplets developed by different teams.

Commercial Transition

Throughout the program, performers will work closely with government, private sector, and clinical partners to refine the commercialization approach for the TA1-TA3 components. We envision the proposing teams will include established manufacturing entities capable of both developing the chiplet platform and scaling production to commercialize the individual components. To this end, proposers should identify their strategy for scaling production and briefly state experience to date in scaling production of sensors or related electronics. To ensure the long-term sustainability and widespread adoption of these chiplet-based biosensing technologies, performers will mature the technologies and components in a way that creates clear incentives for continued investment, development, and deployment. Participants are not expected to form a new company or leave their current positions; instead, during the program they should focus on building the capabilities and partnerships needed to rapidly scale and transition the technology.